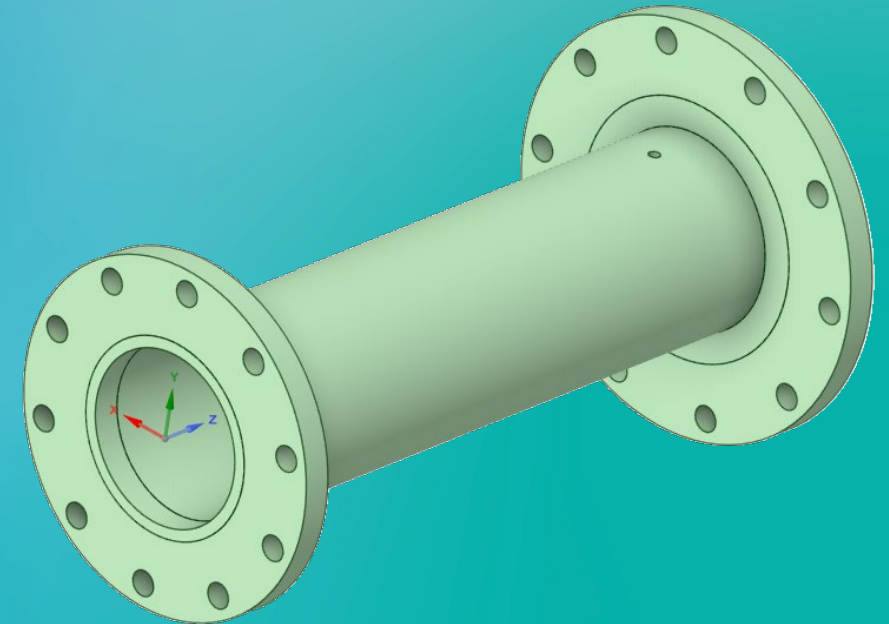


Forschung kompakt:
Windenergie und Wasserstoff
zusammen denken

17.06.2025

Calculation of Adapter to cardan shaft Adapter_Kupplung_20250304.wbpz

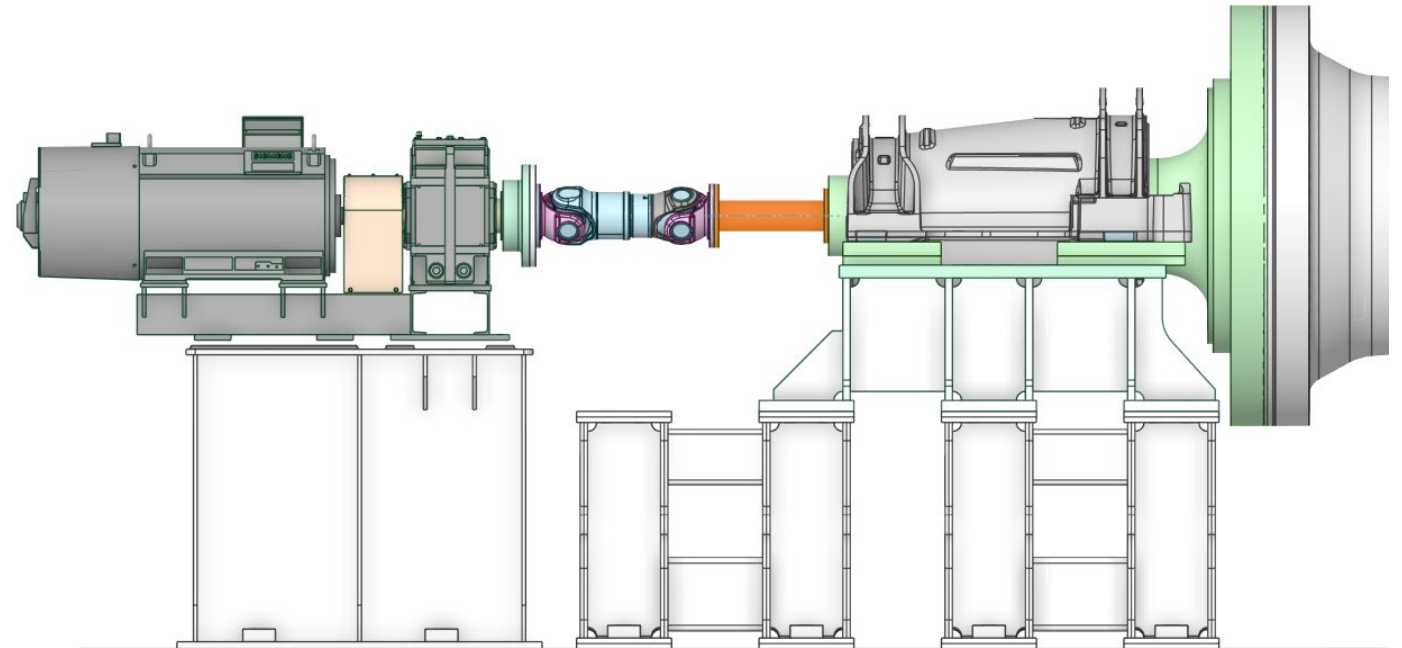
Deepak Petchimuthu



Adapter to cardan shaft

WEA-RiWa testbench setup

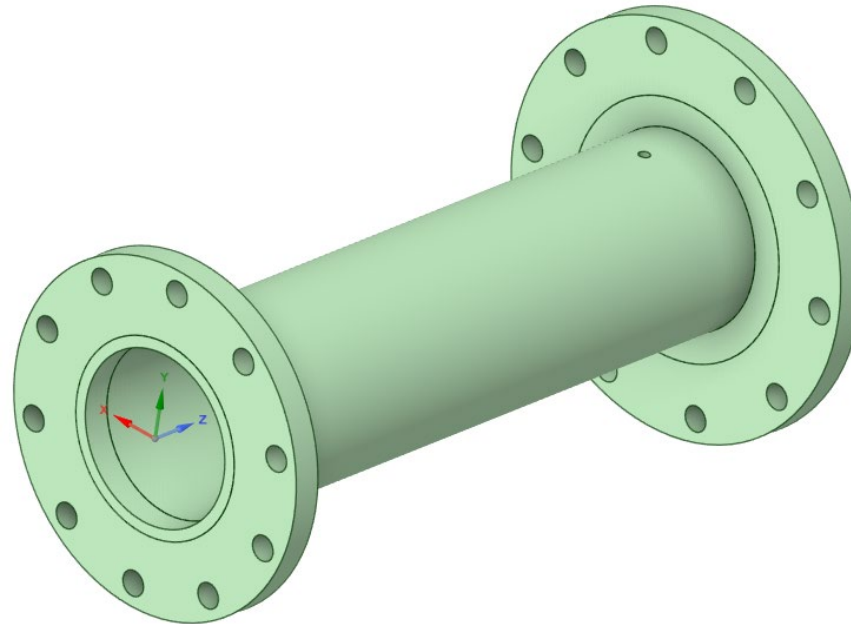
- Connection of cardan shaft to main shaft
- Transfer torque of the drive to the DUT
- Supports the cardan shaft
- Up to 1e6 cycles: 3x DUT's with up to 300000 rotations



Adapter to cardan shaft

Geometry

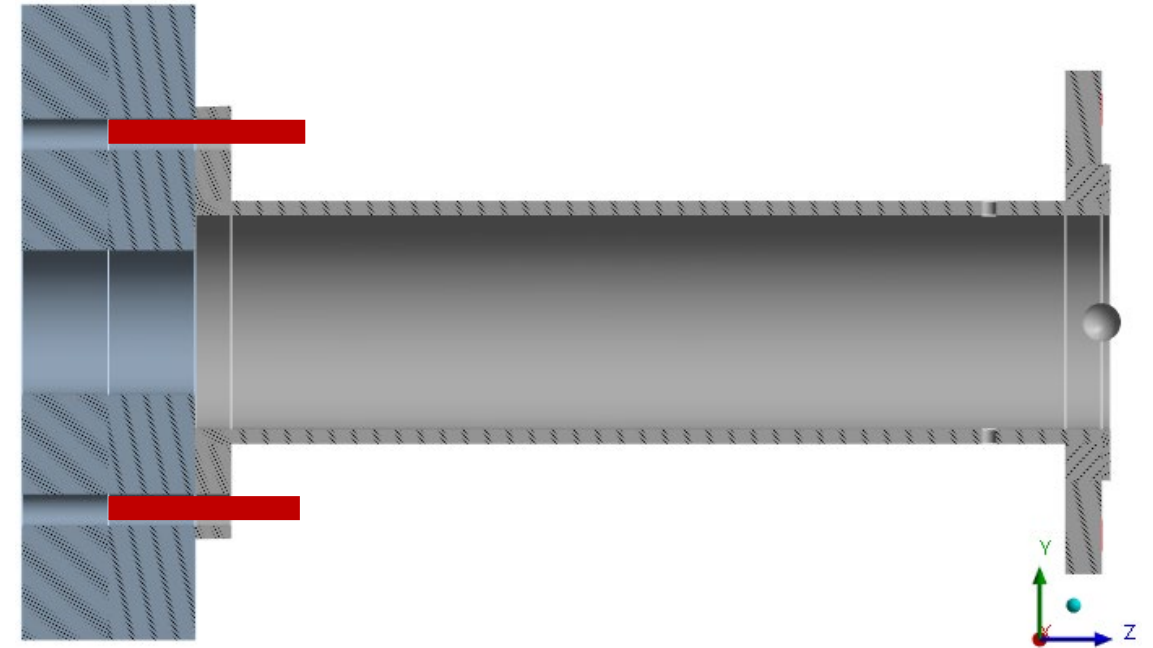
- **Outer tube diameter 168,3 mm**
- **Tube thickness 10 mm**
- **Additional hole dia 10 mm in the tube**



Adapter to cardan shaft

Bolt flange to main shaft

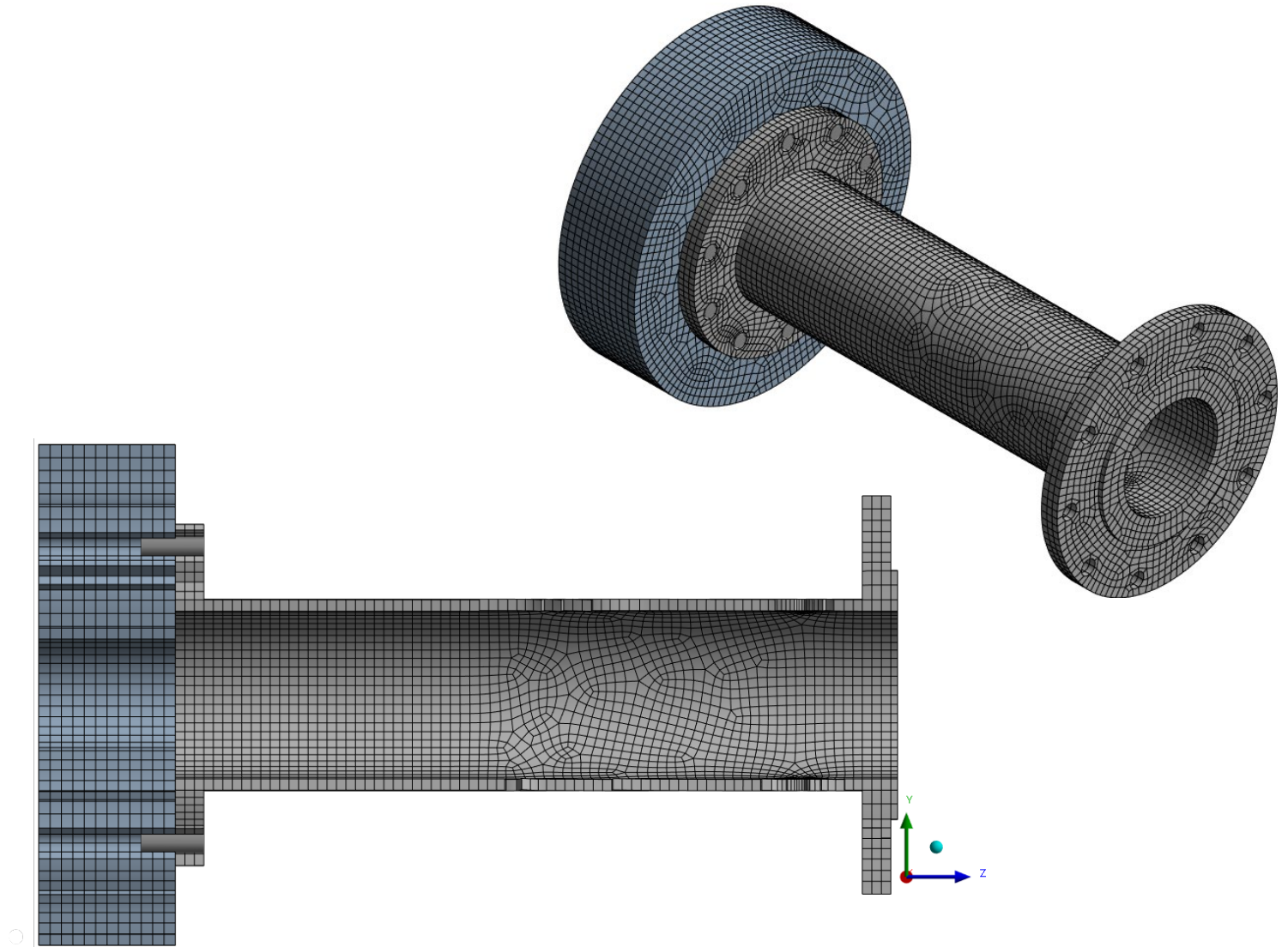
- 10x M20 10.9
- Stress area: $A_s = 244.8 \text{ mm}^2$
Stress diameter: $d_s = 17.65 \text{ mm}$
 $P = 2.5$
- Pretension (90%, $\mu_{K/G}=0,12$) $F_{Mmax} = 185.8 \text{ kN}$
 $F_{Mmin} = 185.8 \text{ kN} / 1.4 = 132.7 \text{ kN}$
- Tightening torque = 592 Nm
- Friction $\mu = 0.2$ (steel to steel)



Adapter to cardan shaft

Mesh

- Hex mesh (quadratic)
- Average element quality: 0.93
- 128485 Nodes / 25127 Elements



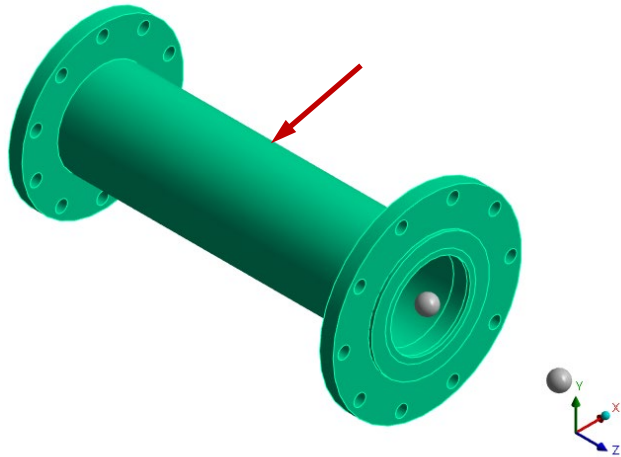
Adapter to cardan shaft

Masses added to the Adapter

- Mass Adapter

→ 47.9 kg

Adapter_Kupplung_simple



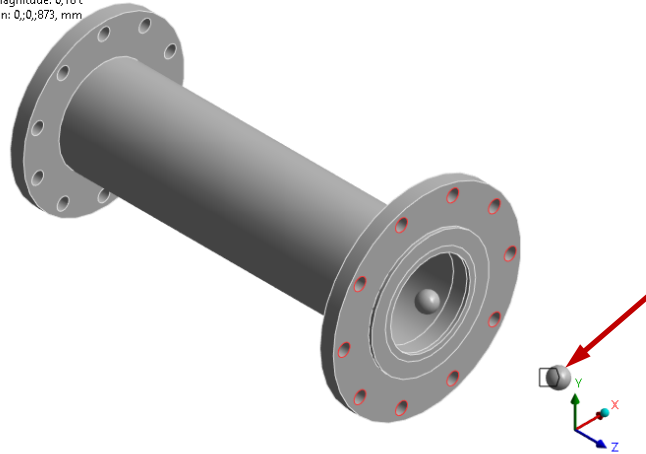
Mass cardan shaft

Half mass of cardan shaft

→ 180 kg

Point Mass cardan shaft

Point Mass cardan shaft
Mass Magnitude: 0,18 t
Location: 0,0,873, mm



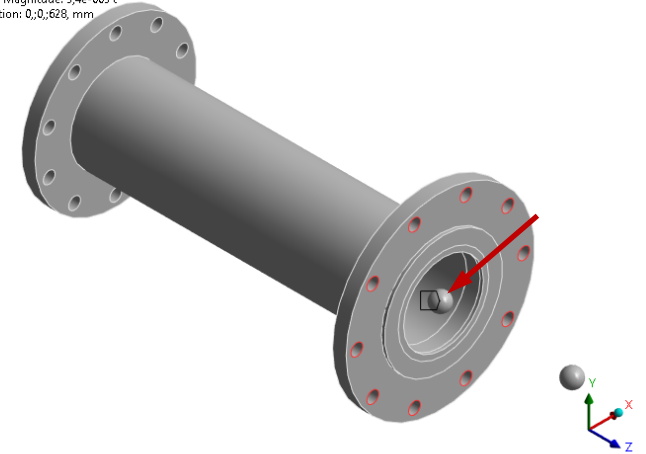
Mass bolts

10x bolts M20x80 & nuts + 20x washer

→ 3.4 kg

Point Mass bolts

Point Mass bolts
Mass Magnitude: 3,4e-003 t
Location: 0,0,628, mm



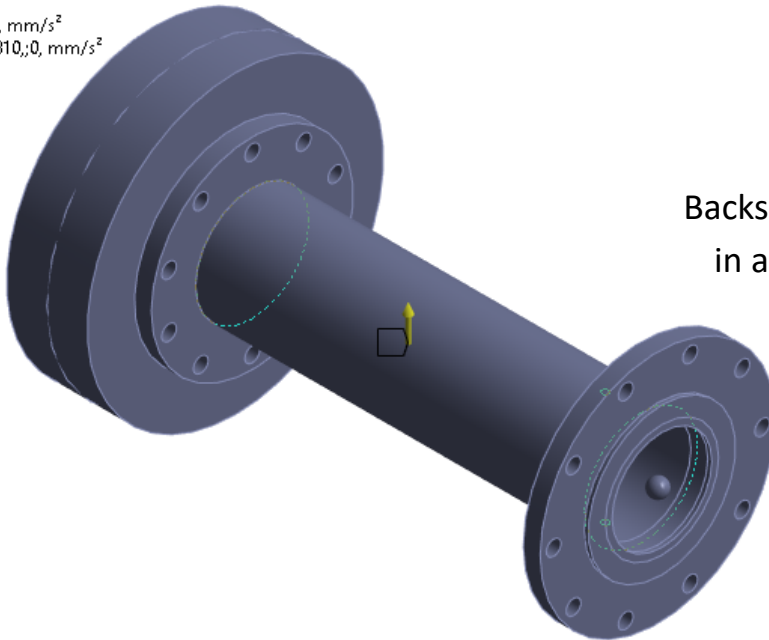
Adapter to cardan shaft

Boundary conditions

- Fatigue case: LC 2 positive and

A: FMmax
Acceleration
Time: 2, s

Acceleration: 9810, mm/s²
Components: 0, 9810, 0, mm/s²

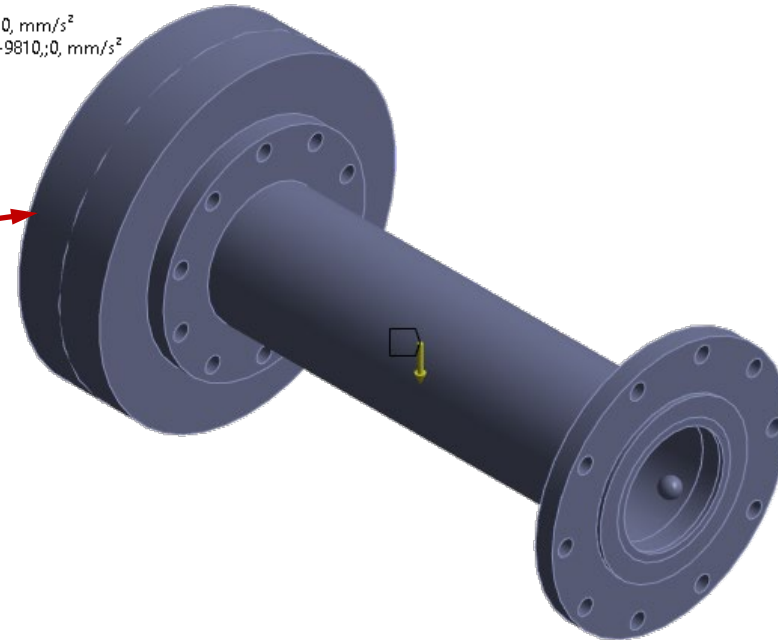


- LC 3 negative self weight

A: FMmax
Acceleration
Time: 3, s

Acceleration: 9810, mm/s²
Components: 0, -9810, 0, mm/s²

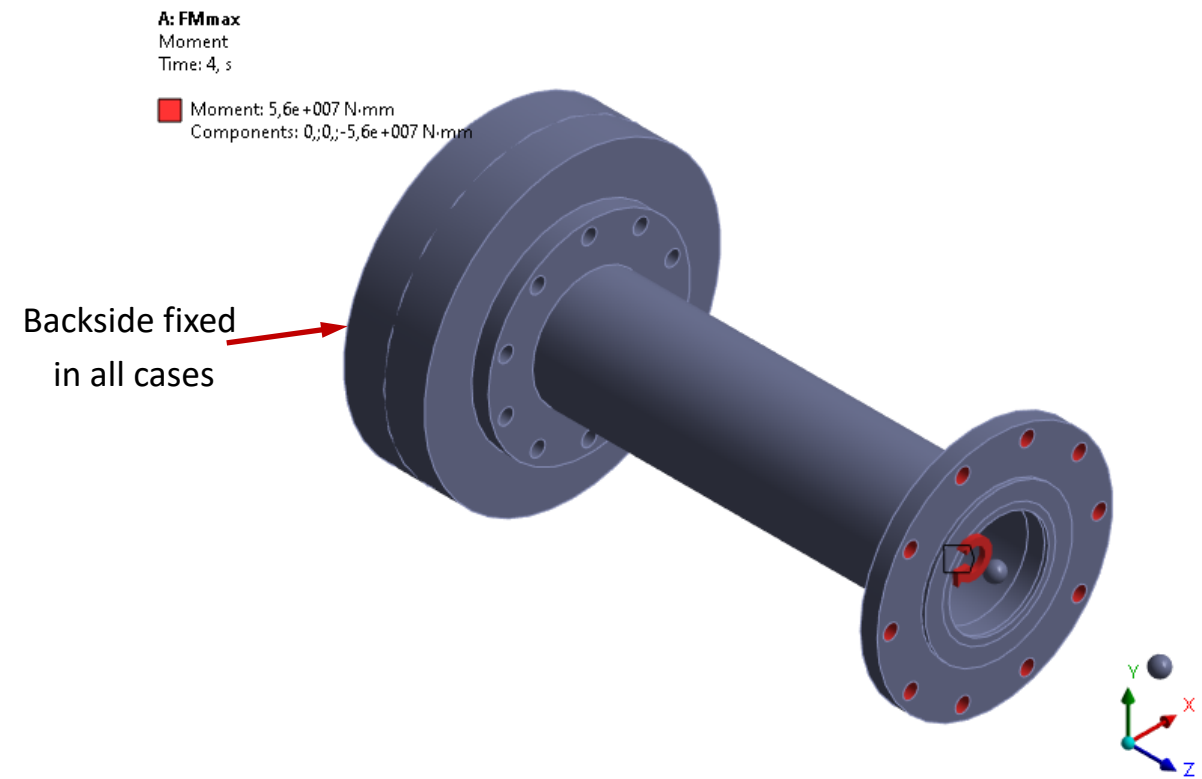
Backside fixed
in all cases



Adapter to cardan shaft

Boundary conditions

- Extreme case: LC 4 Max. Torque load 56 kNm

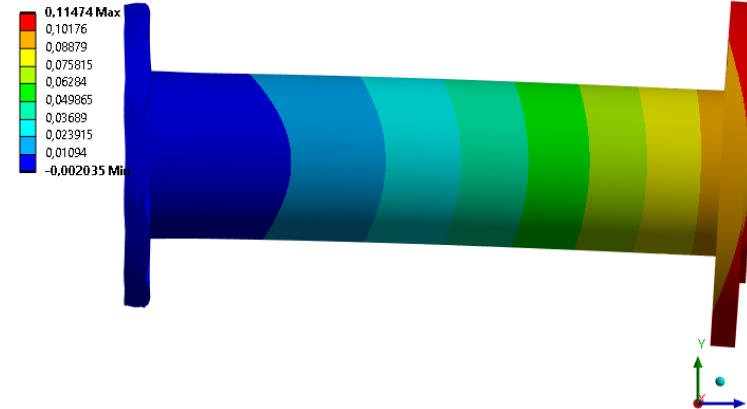


Adapter to cardan shaft

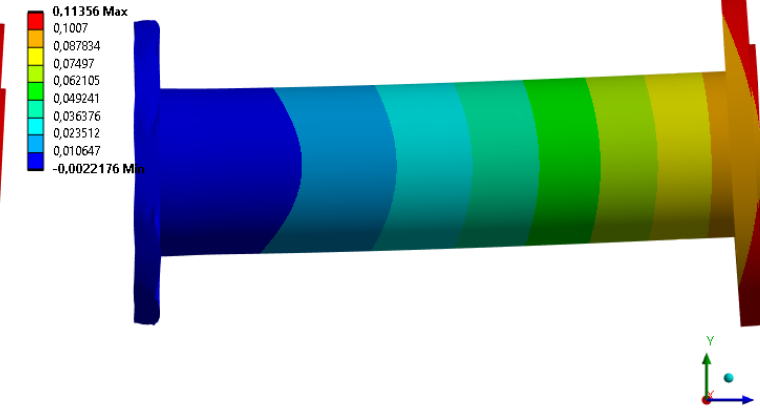
Fatigue Results LC 2 & 3

- Deformation (pretension subtracted)

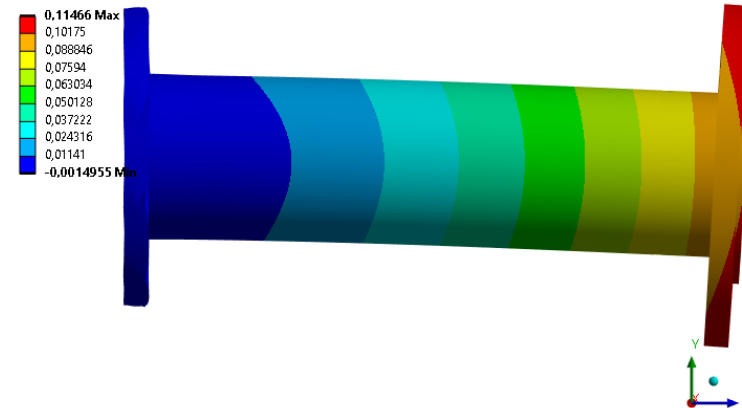
A: FMmax
Adapter_Amplitude_Deformation 2
Expression: D2-D1
Unit: mm
Deformation Scale Factor: 200,



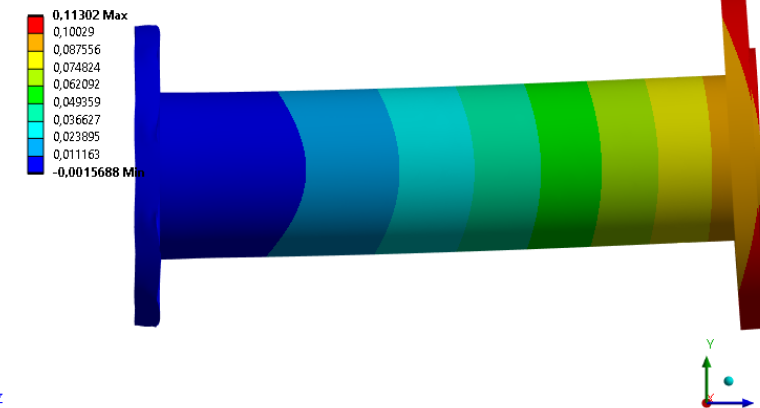
A: FMmax
Adapter_Amplitude_Deformation 3
Expression: D3-D1
Unit: mm
Deformation Scale Factor: 200,



B: FMmin
Adapter_Amplitude_Deformation
Expression: D2min-D1min
Unit: mm
Deformation Scale Factor: 200,



B: FMmin
Adapter_Amplitude_Deformation 2
Expression: D3min-D1min
Unit: mm
Deformation Scale Factor: 200,

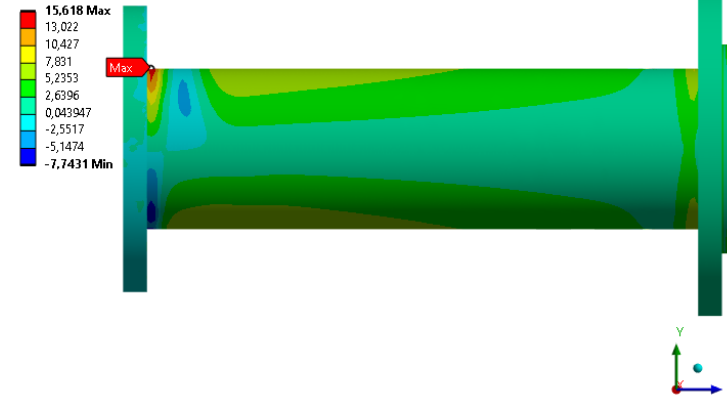


Adapter to cardan shaft

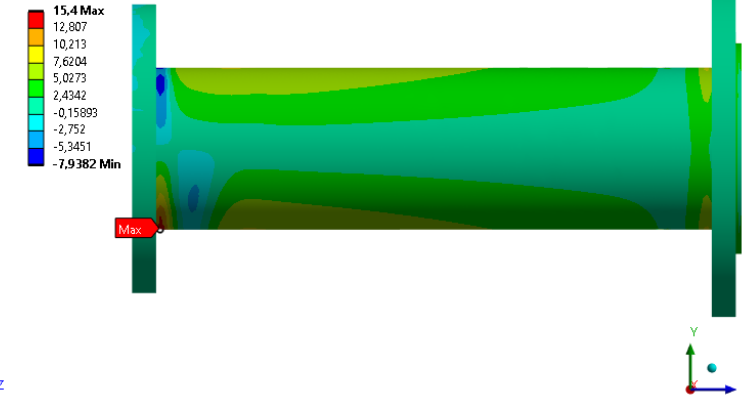
Fatigue Results LC 2 & 3

- vonMises stress (pretension subtracted)

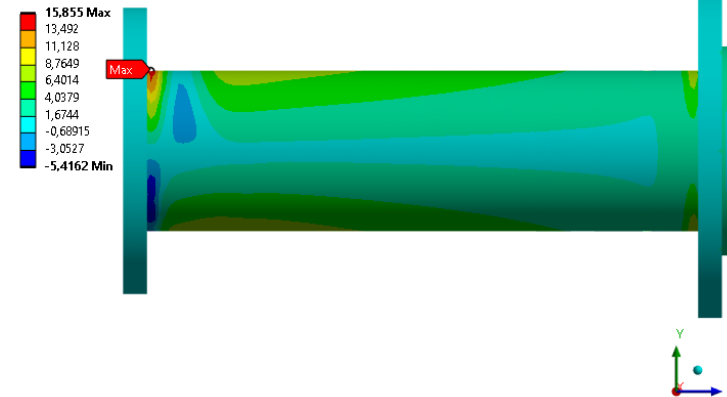
A: FMmax
Adapter_Amplitude 2
Expression: Sa2 = SEQV2-SEQV1
Unit: MPa
Deformation Scale Factor: 1.0 (True Scale)



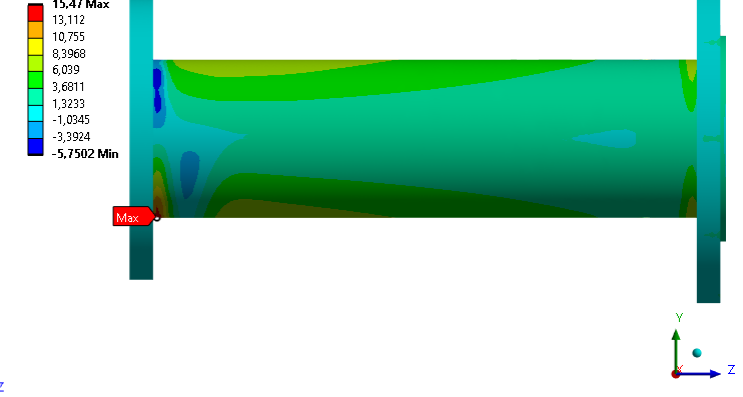
A: FMmax
Adapter_Amplitude 3
Expression: Sa3 = SEQV3-SEQV1
Unit: MPa
Deformation Scale Factor: 1.0 (True Scale)



B: FMmin
Adapter_Amplitude
Expression: SEQV2min-SEQV1min
Unit: MPa
Deformation Scale Factor: 1.0 (True Scale)



B: FMmin
Adapter_Amplitude 2
Expression: SEQV3min-SEQV1min
Unit: MPa
Deformation Scale Factor: 1.0 (True Scale)

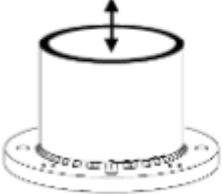
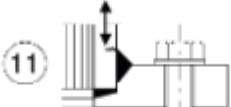
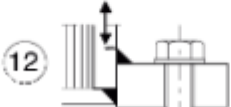


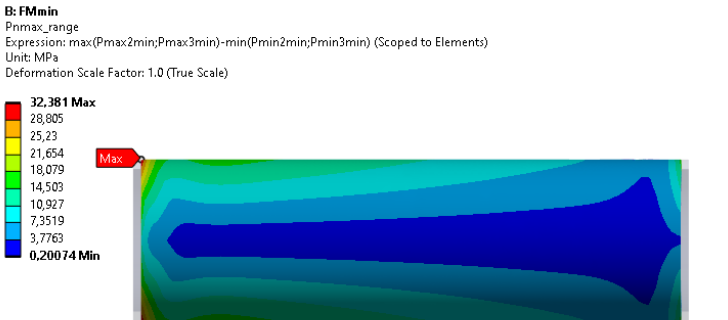
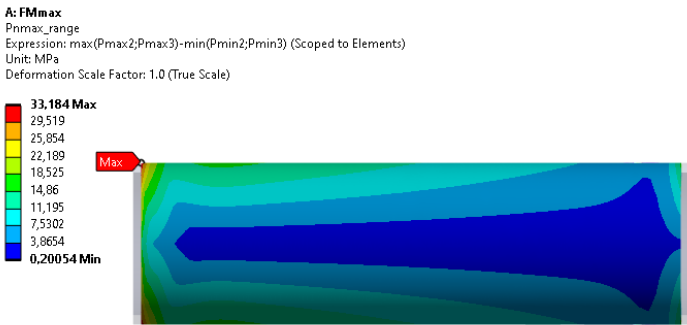
Adapter to cardan shaft

Fatigue Results

- Principal stress range at tube (LCmax - LCmin)

→ 33.2MPa < 40 MPa (@2E6 cycles) acc. to DIN EN 1993-1-9:2010-12 table 8.5

71			11) Tube socket joint with 80% full penetration butt welds.	11) Weld toe ground. $\Delta\sigma$ computed in tube.
40			12) Tube socket joint with fillet welds.	12) $\Delta\sigma$ computed in tube.

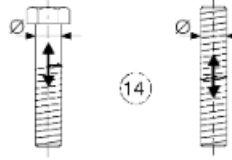


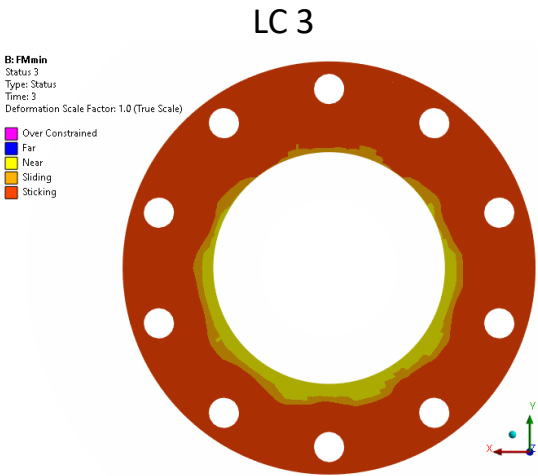
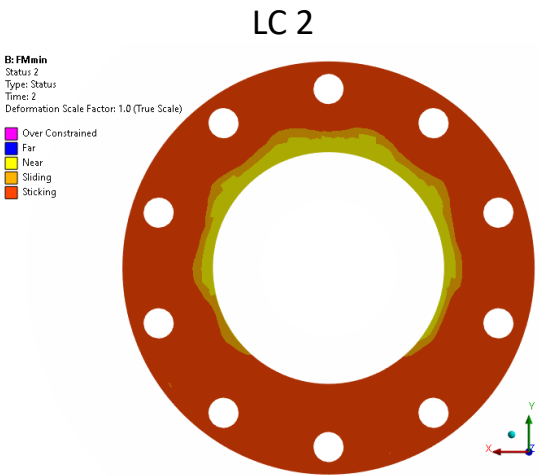
Adapter to cardan shaft

Fatigue Results

- Bolt stress range (LC3 - LC2)

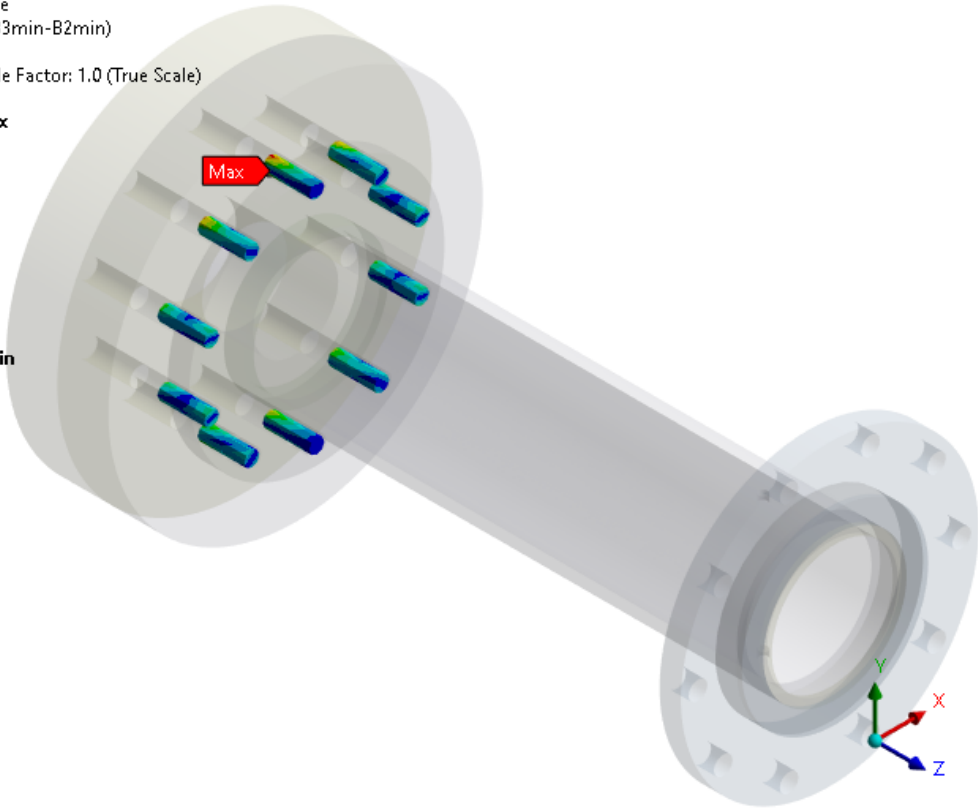
→ 1MPa < 50 MPa (@2E6 cycles) acc. to DIN EN 1993-1-9:2010-12 table 8.1

50	size effect for $\phi > 30\text{mm}$: $k_s = (30/\phi)^{0.23}$		14) Bolts and rods with rolled or cut threads in tension. For large diameters (anchor bolts) the size effect has to be taken into account with k_s .	14) $\Delta\sigma$ to be calculated using the tensile stress area of the bolt. Bending and tension resulting from prying effects and bending stresses from other sources must be taken into account. For preloaded bolts, the reduction of the stress range may be taken into account.
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B: FMmin
Bolt_fatigue range
Expression: abs(B3min-B2min)
Unit: MPa
Deformation Scale Factor: 1.0 (True Scale)

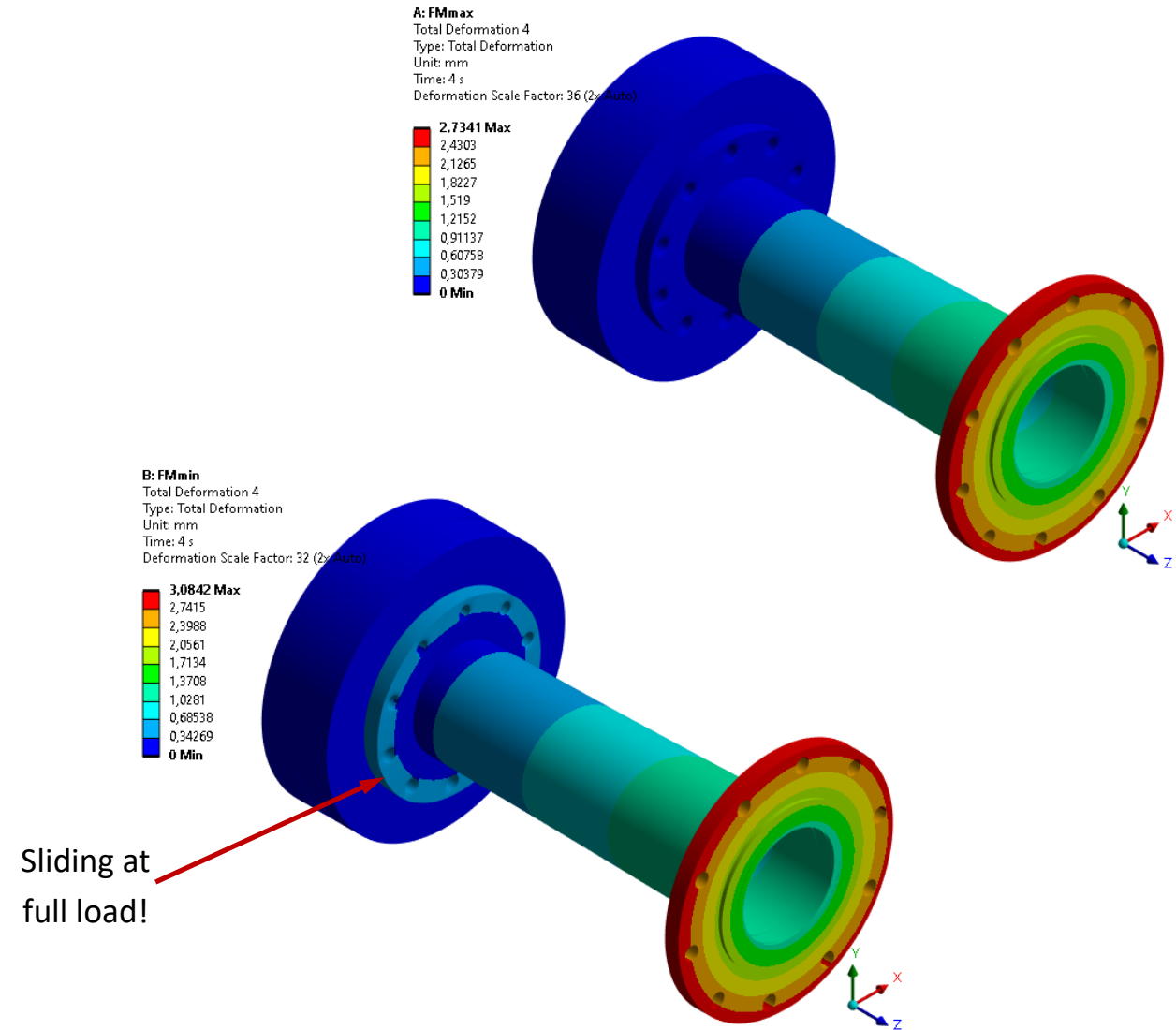
0,88806 Max
0,78941
0,69076
0,59211
0,49345
0,3948
0,29615
0,1975
0,098844
0,000192 Min



Adapter to cardan shaft

Extreme Results LC 4

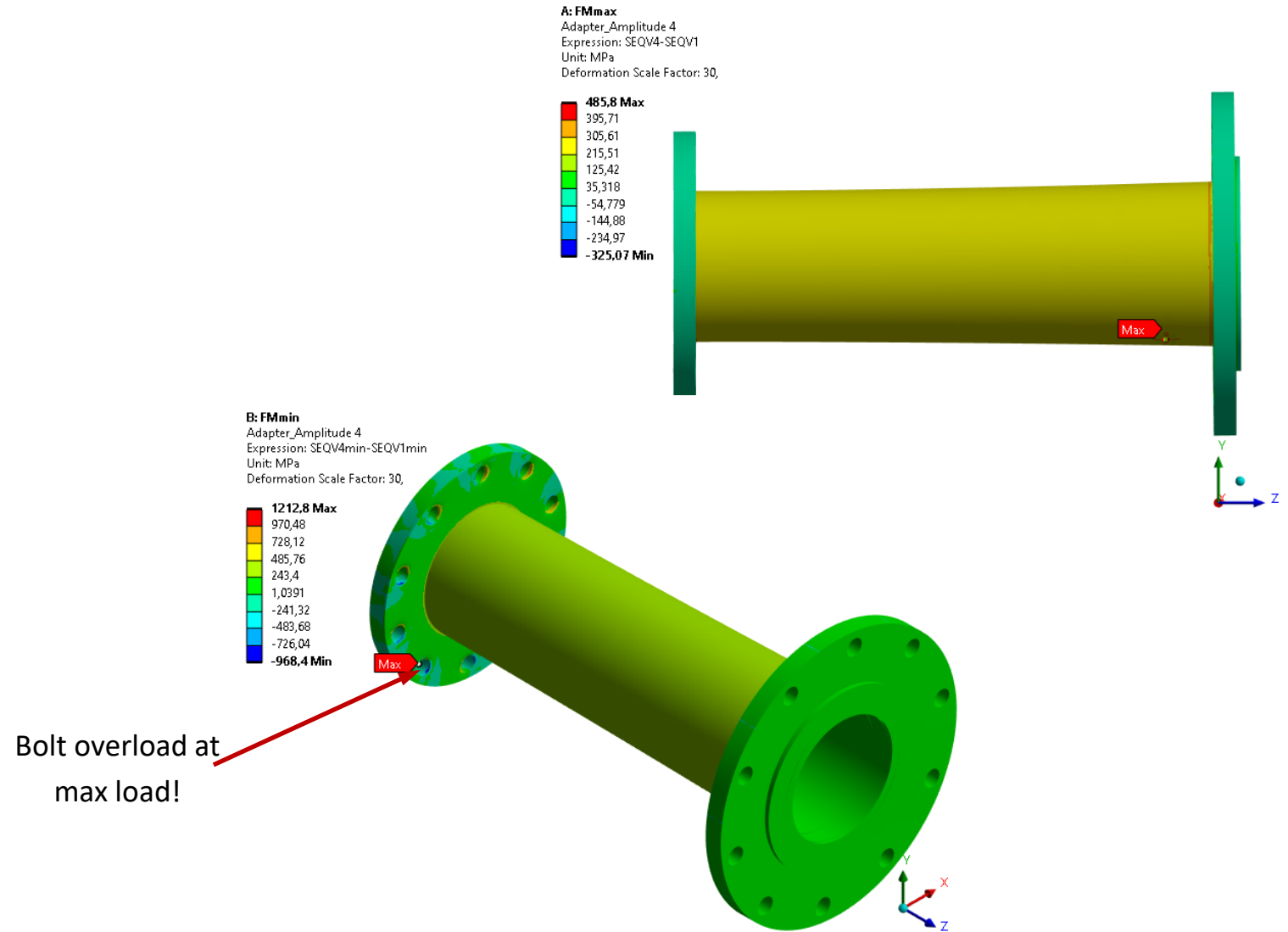
- Deformation



Adapter to cardan shaft

Extreme Results LC 4

- vonMises stress (pretension subtracted)

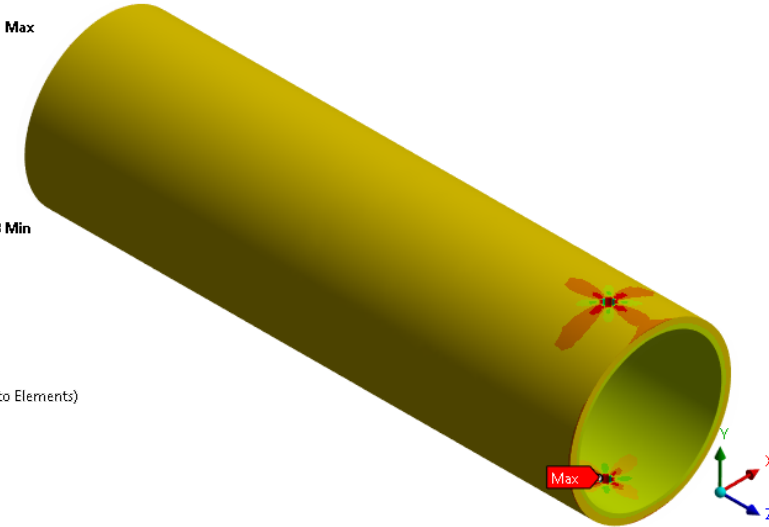
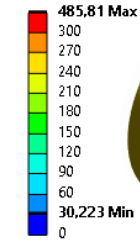


Adapter to cardan shaft

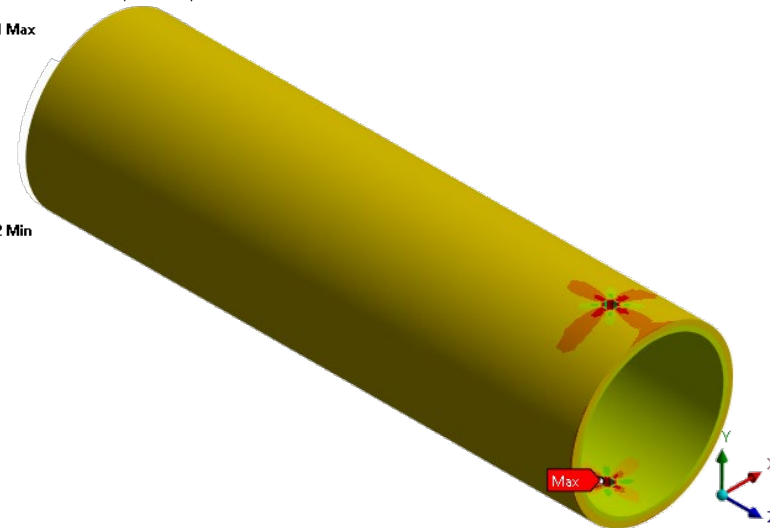
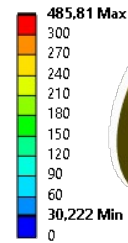
Extreme Results LC 4

- **vonMises stress (pretension subtracted)**
 - tube stress at 262 MPa < 300 MPa (S355)
 - Holes in the tube are not recommended

A: FMmax
Equivalent (von-Mises) Stress - Pipe_reduced - 4, s
Type: Equivalent (von-Mises) Stress (Average Across Bodies) (Scoped to Elements)
Unit: MPa
Time: 4 s
Deformation Scale Factor: 1.0 (True Scale)



B: FMmin
Equivalent (von-Mises) Stress - Pipe_reduced - 4, s
Type: Equivalent (von-Mises) Stress (Average Across Bodies) (Scoped to Elements)
Unit: MPa
Time: 4 s
Deformation Scale Factor: 1.0 (True Scale)



Adapter to cardan shaft

Extreme Results LC 4

- Bolt stress range (LC4) < 940 MPa?

Bolts are overloaded at full load because of sliding in contact

- Analytical check against sliding

Minimum pretension $FM_{min} = 185.8\text{kN} / 1.4 = 132.7\text{kN}$ per bolt

Friction coefficient $\mu = 0.2$

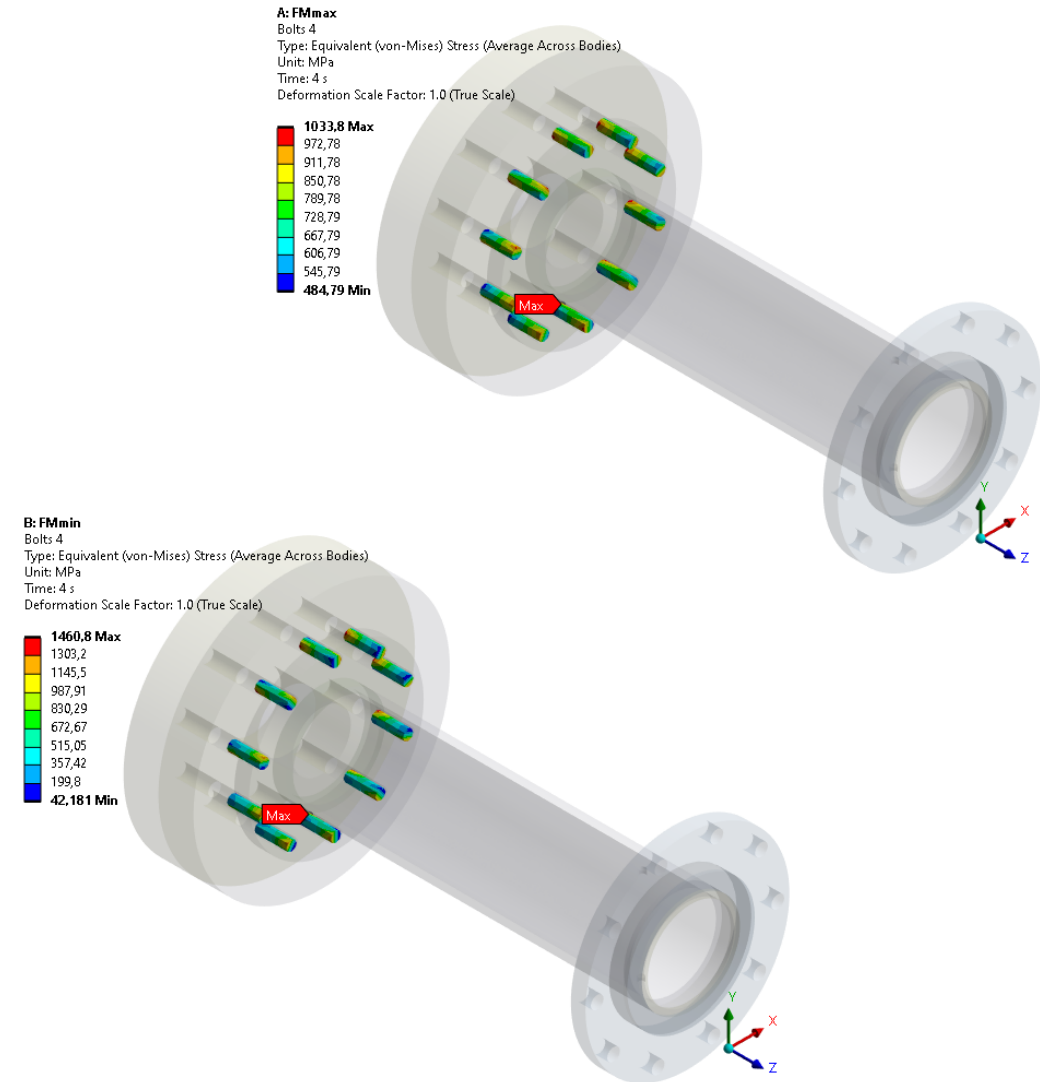
Bolt circle radius $r_b = 130\text{mm}$

Number of bolts $n_b = 10$

Max. Torque $T_{max} = FM_{min} * \mu * n_b * r_b$

Max. Torque $T_{max} = 132.7\text{kN} * 0.2 * 10 * 0.13\text{m}$

Max. Torque $T_{max} = 34.5\text{ kNm}$



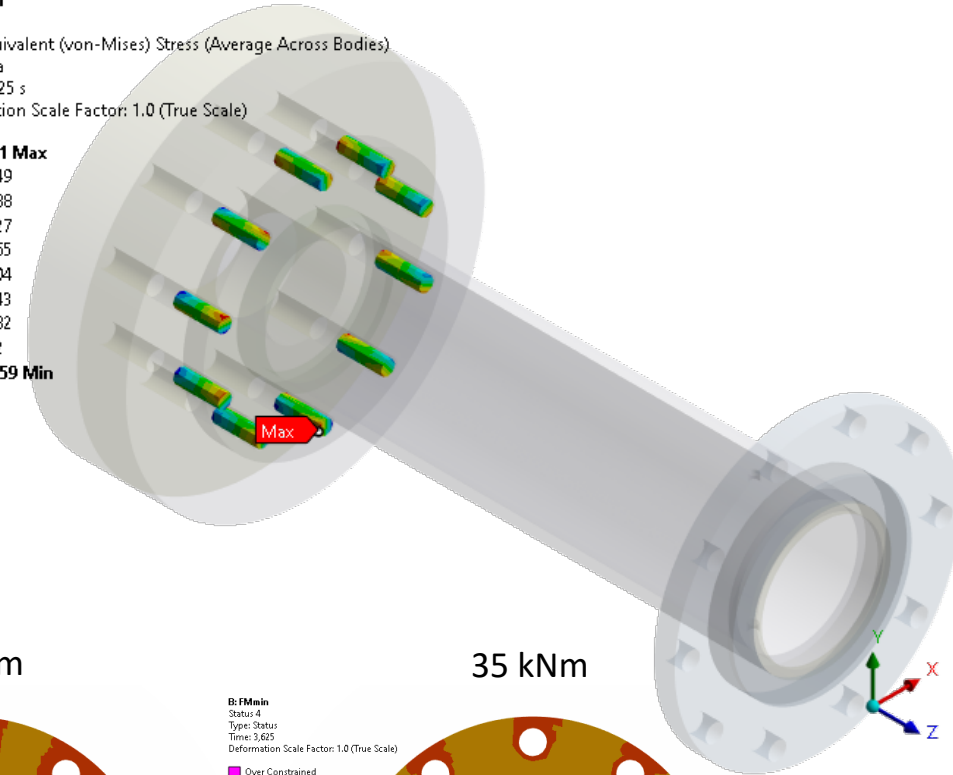
Adapter to cardan shaft

Extreme: Applicable Torque are limited due to bolts

- **Bolt stress range (LC4) < 940 MPa?**
→ up to 35 kNm no sliding expected
- **Contact Status Sticking?**
→ up to 35 kNm sticking around the bolts
→ Max. Torque load of 56 kNm can damage the bolts

B: FMmin
Bolts 4
Type: Equivalent (von-Mises) Stress (Average Across Bodies)
Unit: MPa
Time: 3,625 s
Deformation Scale Factor: 1.0 (True Scale)

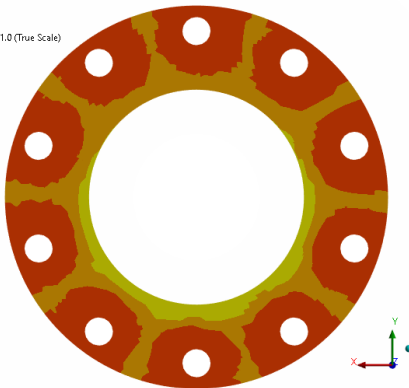
608,1 Max
593,49
578,88
564,27
549,65
535,04
520,43
505,82
491,2
476,59 Min



Torque: 14 kNm

B: FMmin
Status 4
Type: Status
Time: 3,25
Deformation Scale Factor: 1.0 (True Scale)

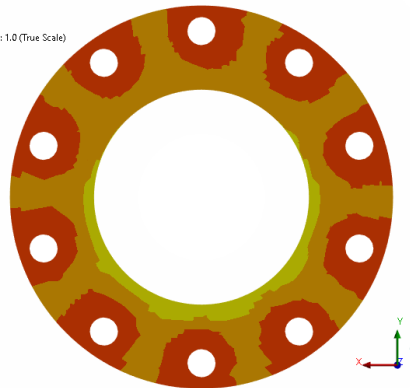
Over Constrained
Far
Near
Sliding
Sticking



21 kNm

B: FMmin
Status 4
Type: Status
Time: 3,375
Deformation Scale Factor: 1.0 (True Scale)

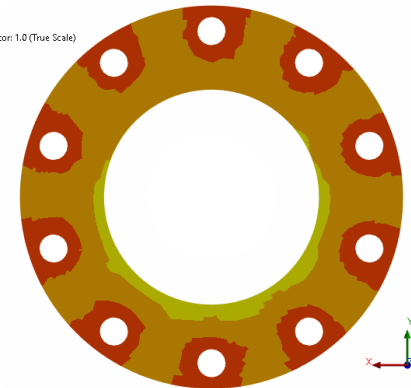
Over Constrained
Far
Near
Sliding
Sticking



28 kNm

B: FMmin
Status 4
Type: Status
Time: 3,5
Deformation Scale Factor: 1.0 (True Scale)

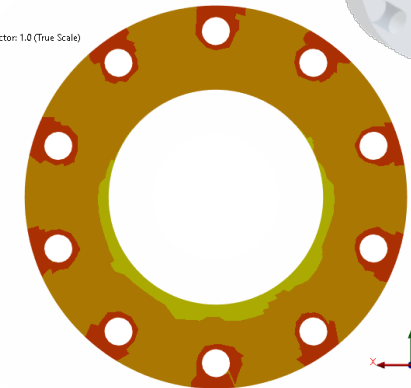
Over Constrained
Far
Near
Sliding
Sticking



35 kNm

B: FMmin
Status 4
Type: Status
Time: 3,625
Deformation Scale Factor: 1.0 (True Scale)

Over Constrained
Far
Near
Sliding
Sticking

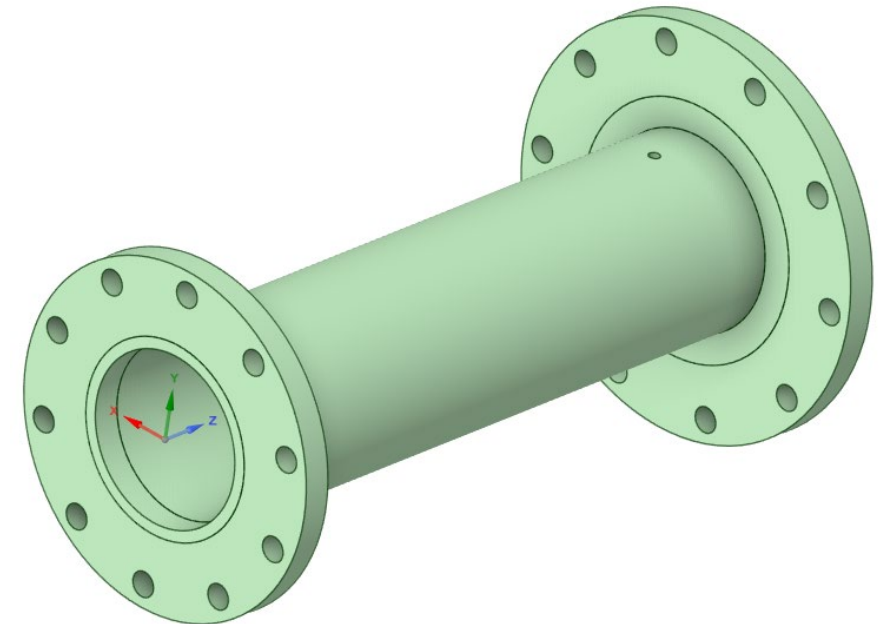




Adapter to cardan shaft

Conclusion

- **Fatigue utilization for 2e6 cycles:**
 - tube/weld stress range utilization → 83%
 - Bolt stress range → 2%
 - **Extreme (max. Torque = 56 kNm) utilization:**
 - Tube/weld stress → 87% (without holes)
 - Bolt stress range → 155% → Torquelimit to 35 kNm
 - Bolt contact flange sliding → ≈160% → Torquelimit to 35 kNm
- 
- **Extreme (Torquelimit = 35 kNm) utilization:**
 - Tube/weld stress → 87% (without holes)
 - Bolt stress range → ≈100%
 - Bolt contact flange sliding → ≈100%





Vielen Dank für Ihre
Aufmerksamkeit!
